

A One-Dimensional Obstruction to Unweighted Sampling of Continuous Schauder Frames

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Status

This packet gives a counterexample to Problem 6.1 in Eisner–Freeman, arXiv:1812.08360 [1], under the literal unweighted definition of sampling stated in the source paper. The counterexample is already one-dimensional.

Source Problem

The source paper defines a sampling of a continuous frame $(x_t, f_t)_{t \in M}$ by choosing a sequence $(t_j)_{j=1}^\infty$ in M and considering the discrete sequence $(x_{t_j}, f_{t_j})_{j=1}^\infty$. Problem 6.1 asks which Banach spaces have the property that every bounded continuous Schauder frame may be sampled to obtain a discrete Schauder frame, and which Banach spaces admit a bounded continuous Schauder frame which cannot be so sampled.

Proof Intuition

The obstruction is that the source’s formal sampling operation records only the raw sampled pairs. It does not attach quadrature weights. Thus a continuous identity decomposition over a finite non-atomic measure space can have total mass one, while every infinite unweighted sample repeats the same rank-one identity contribution infinitely many times.

This is invisible in weighted Hilbert-frame discretization, where one may sample with weights. Here, as written, the repeated unweighted coefficient series diverges.

Theorem 1. *The one-dimensional Banach space $\mathbb{R}$ has a bounded continuous Schauder frame which cannot be sampled, in the unweighted sense of [1], to obtain a discrete Schauder frame.*

Proof. Let $X = \mathbb{R}$, let $M = [0, 1]$ with Lebesgue measure, and define

$$x_t = 1 \in \mathbb{R}, \quad f_t = \text{Id}_{\mathbb{R}} \in X^* \quad (t \in [0, 1]).$$

The map $t \mapsto (x_t, f_t)$ is measurable and bounded, with $\|x_t\| \|f_t\| = 1$ for all t .

For every scalar $x \in \mathbb{R}$,

$$\int_0^1 f_t(x) x_t dt = \int_0^1 x dt = x.$$

Moreover, for every measurable set $E \subset [0, 1]$, the Pettis integral over E is the scalar $\mu(E)x$. Hence the source definition of continuous Schauder frame is satisfied.

6. SAMPLING CONTINUOUS SCHAUDER FRAMES

Many important frames for Hilbert spaces arise as samplings of continuous frames. In particular, wavelet frames, Gabor frames, and Fourier frames are all samplings of different continuous frames. Furthermore, all the frames introduced by Daubechies, Grossmann, and Meyer [DGM] in “Painless nonorthogonal expansions” are created by sampling different coherent states. Formally, if (M, Σ, μ) is a σ -finite measure space and $(x_t, f_t)_{t \in M}$ is a continuous frame of a Banach space X and $(t_j)_{j=1}^\infty$ is a sequence in M then $(x_{t_j}, f_{t_j})_{j=1}^\infty$ is called a sampling of $(x_t, f_t)_{t \in M}$. The discretization problem, posed by Ali, Antoine, and Gazeau [AAG2], asks when a continuous frame of a Hilbert space can be sampled to obtain a frame. A solution for certain types of continuous frames was obtained by Fornasier and Rauhut using the theory of co-orbit spaces [FoR] and a complete solution was recently given by Speegle and the second author [FS] using the solution of the Kadison Singer problem by Marcus, Spielman, and Srivastava [MSS]. In particular, every bounded continuous frame on a Hilbert space may be sampled to obtain a discrete frame.

Problem 6.1. What are some Banach spaces where every bounded continuous Schauder frame may be sampled to obtain a discrete Schauder frame? What are some Banach spaces where there exists a bounded continuous Schauder frame which cannot be sampled to obtain a discrete Schauder frame?

Note that the discretization problem was solved for continuous Hilbert space frames, and thus Problem 6.1 is even open for continuous Schauder frames for separable Hilbert spaces.

Figure 1: Problem 6.1 from arXiv:1812.08360, final section.

Now let $(t_j)_{j=1}^\infty$ be any sampling sequence in M . Since the frame is constant in t , the sampled sequence is

$$(x_{t_j}, f_{t_j})_{j=1}^\infty = (1, \text{Id}_{\mathbb{R}})_{j=1}^\infty.$$

If this were a discrete Schauder frame for $\mathbb{R}$, then for $x = 1$ the reconstruction series would have to converge to 1:

$$1 = \sum_{j=1}^\infty f_{t_j}(1)x_{t_j} = \sum_{j=1}^\infty 1.$$

The series on the right diverges. Therefore no sampling sequence produces a discrete Schauder frame. $\square$

Verification Notes

The example is compatible with the source’s continuous Schauder-frame definition, including the requirement that the Pettis integral over every measurable subset exist. For each $E \subset [0, 1]$, the partial integral operator is simply multiplication by $\mu(E)$.

The example is also bounded in the usual sense relevant to the source’s sampling problem: $\sup_t \|x_t\| \|f_t\| = 1$, and the suppression unconditionality constant is 1.

Limitations

This is a counterexample to unweighted sampling exactly as formulated in the source text. It should not be read as a counterexample to weighted discretization of continuous Hilbert frames. In the Hilbert-frame literature, the discretized frame elements are typically allowed to carry sampling weights; with weights, the present one-dimensional example is harmless.

Thus the mathematical conclusion is best phrased as follows: Problem 6.1 has a negative answer under the paper’s literal unweighted definition, and any nontrivial positive discretization problem for continuous Schauder frames must allow weights or impose an additional normalization preventing repeated unit-mass samples.

Novelty and Search Bounds

The local run indexes were searched for arXiv:1812.08360, continuous Schauder frames, sampling, and discretization. Nearby frame packets concern Schauder frame false positives and approximate-frame counterexamples, but no packet for this exact source problem was found.

The source paper itself cites the weighted Hilbert-frame discretization theorem of Freeman–Speegle. A bounded web search on June 21, 2026 found the source paper and adjacent continuous-frame discretization material, but no later Banach-space answer to Problem 6.1 and no note pointing out this unweighted one-dimensional obstruction.

Human Review Recommendation

This packet should be reviewed as a wording-sensitive counterexample. The proof is elementary; the key review question is interpretive: whether Problem 6.1 is to be read literally with unweighted raw sampled pairs, or whether the authors intended weighted sampling by analogy with Hilbert-frame discretization.

References

- [1] J. Eisner and D. Freeman, *Continuous Schauder frames for Banach spaces*, arXiv:1812.08360.